Unit 3:- Differential Calculus, Curve Tracing and Infinite Series:

Topics:

Successive Differentiation: Some standard results, Leibnitz

Problems:

- Example 4, Q. 1 to 19. (page 21, CP)

Topics:

Taylors and Maclaurin series:

Problems:

- Example 5, Q. 1 to 15. (page 26, CP)

Topics:

Asymptotes: Definition, Rules for finding Asymptotes, Two Parallel Asymptotes, Asymptotes Parallel to the x and y axis.

Problems:

- Example 1 to 6 (page 206, GP)
- Example 1 to 6 (Page 209, GP)
- Example 1 to 5 (Page 212, GP)

Topics:

Tracing of Curves.: Procedure for tracing Cartesian Curves, Polar equations

Problems:

- Exercise 13, Q. 1 to 20. (page 70, CP)
- Tracing of Cycloid, Cardioid, Lemiscate, Limacon, and Spiral.

Topics:

Infinite Series: Infinite series, Ratio test, Comparison test and root test of convergence

Problems:

- Exercise 79, Q. 1 to 15 (page 405, CP)
- Exercise 80, Q. 1 to 15 (page 408, CP)

Reference Books

1. (CP)Chandrika Prasad: Mathematics for Engineers
2. (GP)Gorakh Prasad: Differential Calculus.

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